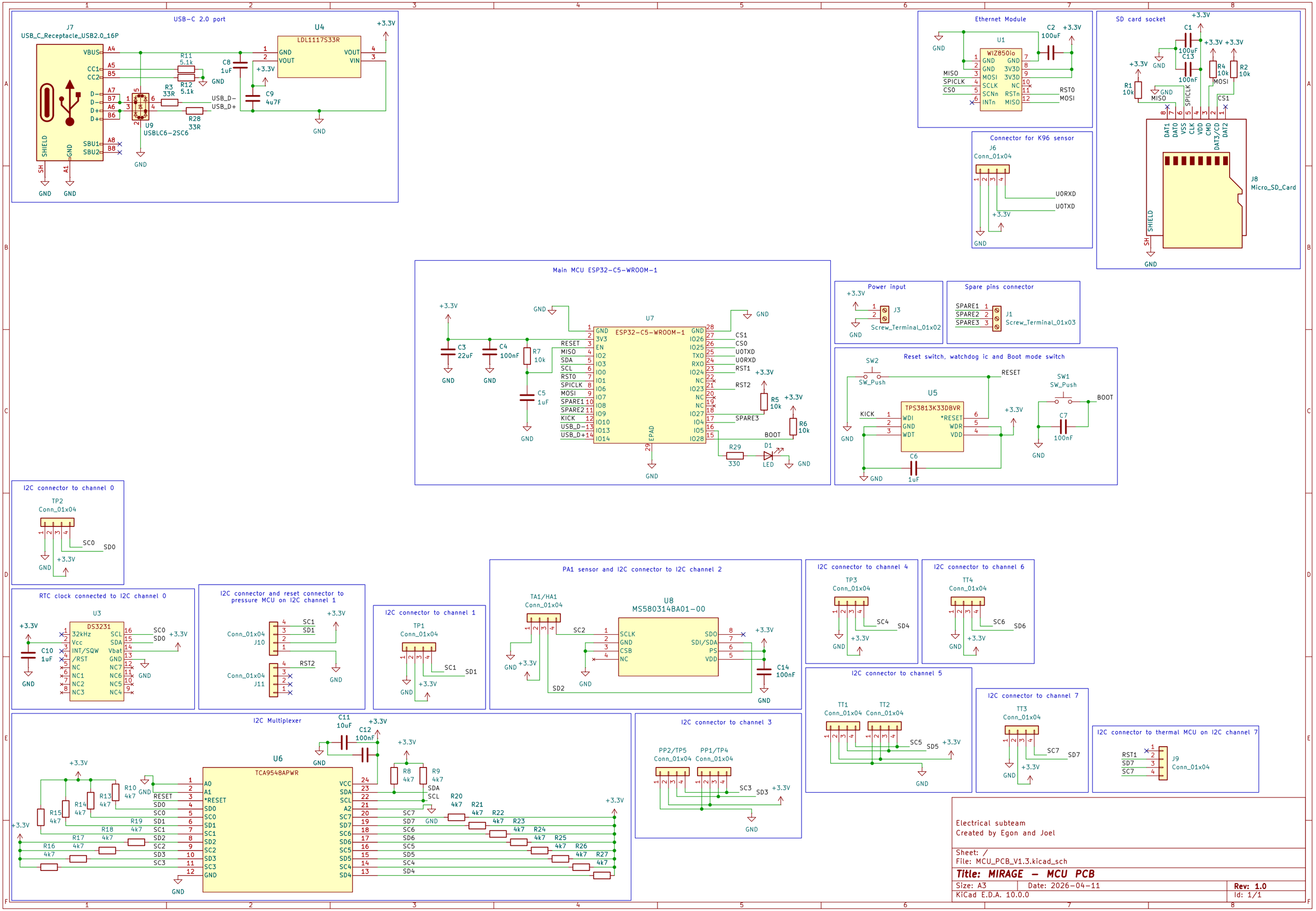
USB-C 2.0 port
J7
USB_C_Receptacle_USB2.0_16P
U4
LDL1117S33R
R11 5.1k
R12 5.1k
R3 33R
R28 33R
C8 1uF
C9 4u7F
U9
USBLC6-2SC6
USB_D-
USB_D+
Ethernet Module
U1
WIZ850io
C2 100uF
Connector for K96 sensor
J6
Conn_01x04
U0RXD
U0TXD
SD card socket
C1 100uF
C13 100nF
R4 10k
R2 10k
R1 10k
J8
Micro_SD_Card
Main MCU ESP32-C5-WROOM-1
U7
ESP32-C5-WROOM-1
C3 22uF
C4 100nF
R7 10k
C5 1uF
R5 10k
R6 10k
R29 330
D1
LED
Power input
J3
Screw_Terminal_01x02
Spare pins connector
J1
Screw_Terminal_01x03
Reset switch, watchdog ic and Boot mode switch
SW2
SW_Push
U5
TPS3813K33DBVR
SW1
C7 100nF
C6 1uF
I2C connector to channel 0
TP2
Conn_01x04
RTC clock connected to I2C channel 0
U3
DS3231
C10 1uF
I2C connector and reset connector to pressure MCU on I2C channel 1
Conn_01x04
J10
J11
I2C connector to channel 1
TP1
PA1 sensor and I2C connector to I2C channel 2
TA1/HA1
U8
MS580314BA01-00
C14 100nF
I2C connector to channel 4
TP3
I2C connector to channel 6
TT4
I2C connector to channel 5
TT1
TT2
I2C connector to channel 7
TT3
I2C connector to thermal MCU on I2C channel 7
J9
I2C Multiplexer
U6
TCA9548APWR
C11 10uF
C12 100nF
R8 4k7
R9 4k7
R10 4k7
R13 4k7
R14 4k7
R15 4k7
R16 4k7
R17 4k7
R18 4k7
R19 4k7
R20 4k7
R21 4k7
R22 4k7
R23 4k7
R24 4k7
R25 4k7
R26 4k7
R27 4k7
I2C connector to channel 3
PP2/TP5
PP1/TP4
Electrical subteam
Created by Egon and Joel
Sheet: /
File: MCU_PCB_V1.3.kicad_sch
Title: MIRAGE - MCU PCB
Size: A3
Date: 2026-04-11
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 1/1